



Dunnhumby Retail Analytics

Solving real business problems with SQL



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Business Problem

A grocery retail chain wants to understand **what drives customer spending**, which product categories are most profitable, how effective **promotional campaigns and coupons** are in influencing purchase behavior, and which **customer segments** contribute most to overall revenue — with the goal of optimizing promotions, product assortment, and pricing strategy.

Revenue

Which depts, stores and brands drive the most value?

Promotions

Do coupons and campaigns actually lift sales?

Customers

Who are the high-value shoppers and what do they buy?

Dataset & Schema

transaction_data

~ 2.5M records

COLUMNS

HOUSEHOLD_KEY	BASKET_ID	DAY	PRODUCT_ID	QUANTITY
SALES_VALUE	STORE_ID	WEEK_NO	RETAIL_DISC	COUPON_DISC

product

~ 92K products

PRODUCT_ID	DEPARTMENT	BRAND	COMMODITY_DESC
SUB_COMMODITY_DESC			

coupon

Campaign mapping

COUPON_UPC	PRODUCT_ID	CAMPAIGN
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All 3 tables joined via PRODUCT_ID — enabling cross-dataset analysis

Identifies which departments generate the most revenue — guides product assortment and shelf-space decisions.

```
select p.department,  
       round(sum(t.sales_value), 2) total_revenue,  
       count(distinct t.basket_id) total_transactions,  
       round(avg(t.sales_value), 2) avg_sale_per_item  
from transaction_data t  
join product p on t.product_id = p.product_id  
group by p.department  
order by total_revenue desc  
limit 10;
```

department	total_revenue	total_transactions	avg_sale_per_item
abc Filter...	abc Filter...	abc Filter...	abc Filter...
GROCERY	4093814.14	215411	2.49
DRUG GM	1055358.03	118195	3.81
PRODUCE	557452.11	89026	2.17
MEAT	548786.81	53401	6.21
KIOSK-GAS	544222.28	22056	24.67
MEAT-PCKGD	412436.77	58466	3.68
DELI	260866.51	35639	4.15
PASTRY	121739.86	30099	3.19
MISC SALES TRAN	119960.04	5992	19.83
NUTRITION	97669.04	16820	3.04

Shows where the store is giving away the most margin — useful for renegotiating supplier terms or adjusting discount strategy.

```
select
p.COMMODITY_DESC,
round(sum(t.retail_disc), 2)    total_retail_discount,
round(sum(t.sales_value), 2)    total_sales,
round(sum(t.retail_disc) / nullif(sum(t.sales_value), 0) * 100, 2)    discount_pct
from transaction_data t
join product p on t.product_id = p.product_id
group by p.commodity_desc
having sum(t.sales_value) > 0
order by discount_pct desc
limit 15;
```

COMMODITY_D...	total_retail_disc...	total_sales	discount_pct
abc Filter...	abc Filter...	abc Filter...	abc Filter...
(CORP USE ONLY)	0.00	283.40	0.00
BOTTLE DEPOSITS	0.00	116.60	0.00
BOUQUET (NON ROSE)	0.00	80.90	0.00
BULK FOODS	0.00	41.02	0.00
CIGARS	0.00	668.75	0.00
COFFEE SHOP SWEET ...	0.00	492.37	0.00
COUPON	0.00	459.66	0.00
DELI SUPPLIES	0.00	1965.87	0.00
EASTER LILY	0.00	31.98	0.00
FROZEN PACKAGE ME...	0.00	12.99	0.00
FRZN ICE	0.00	6576.25	0.00
GIFT & FRUIT BASKETS	0.00	599.87	0.00
GLASSES/VISION AIDS	0.00	1276.18	0.00
HOME HEALTH CARE	0.00	126.07	0.00
MEAT SUPPLIES	0.00	273.59	0.00

Reveals seasonal or promotional revenue cycles across the full dataset period

```
select
week_no,
round(sum(sales_value), 2)    weekly_revenue,
count(distinct basket_id)    weekly_transactions,
count(distinct household_key) unique_shoppers
from transaction_data
group by week_no
order by week_no;
```

week_no	weekly_revenue	weekly_transacti...	unique_shoppers
 Filter...	 Filter...	 Filter...	 Filter...
1	5211.16	142	88
2	10821.35	324	175
3	13498.20	482	228
4	15965.99	545	270
5	20139.82	757	370
6	24923.90	861	433
7	28073.97	999	491
8	31793.77	1086	530
9	31244.17	1212	622
10	42994.38	1553	708
11	51011.30	1823	846
12	53187.30	1926	905
13	55203.01	2203	972
14	62875.20	2433	1119
15	74261.29	2743	1156
16	79135.01	2840	1309
17	79556.35	3048	1371
18	75968.18	3080	1313
19	72536.88	2973	1268
20	86108.90	3091	1319

Shows which departments are most coupon-dependent — helps evaluate promotional dependency by category

```
select p.department,
round(sum(t.sales_value),2) total_sales,
round(sum(case when t.coupon_disc > 0
|         | then t.sales_value else 0 end), 2) coupon_sales,
round(sum(case when t.coupon_disc > 0
|         | then t.sales_value else 0 end)* 100.0 / nullif(sum(t.sales_value), 0), 2) coupon_sales_pct
from transaction_data t
join product p on t.product_id = p.product_id
group by p.department
order by coupon_sales_pct desc;
```

department	total_sales	coupon_sales	coupon_sales_pct
 Filter...	 Filter...	 Filter...	 Filter...
COSMETICS	32360.37	0.00	0.00
AUTOMOTIVE	452.66	0.00	0.00
CHARITABLE CONT	7.74	0.00	0.00
CHEF SHOPPE	2290.80	0.00	0.00
CNTRL/STORE SUP	49.05	0.00	0.00
DRUG GM	1055358.03	0.00	0.00
COUP/STR & MFG	1027.72	0.00	0.00
DAIRY DELI	73.23	0.00	0.00
DELI	260866.51	0.00	0.00
DELI/SNACK BAR	36.08	0.00	0.00
VIDEO RENTAL	21.46	0.00	0.00
ELECT & PLUMBING	1.00	0.00	0.00
FLORAL	39653.34	0.00	0.00
FROZEN GROCERY	653.26	0.00	0.00
GARDEN CENTER	7667.82	0.00	0.00
GM MERCH EXP	101.10	0.00	0.00
GRO BAKERY	2.18	0.00	0.00
GROCERY	4093814.14	0.00	0.00
HBC	9.42	0.00	0.00

Identifies loyal, habitual customers per product — a signal for subscription or loyalty reward targeting

```
select
  household_key,
  product_id,
  count(*) as purchase_count,
  round(sum(sales_value), 2) as total_spent_on_product
from transaction_data
group by household_key, product_id
having count(*) >= 3
order by purchase_count desc
limit 50;
```

household_key	product_id	purchase_count	total_spent_on...
 Filter...	 Filter...	 Filter...	 Filter...
1510	1126203	246	123.50
1510	894236	240	123.50
2337	1120741	240	145.29
1023	6533889	236	9444.06
853	975938	220	223.78
900	890536	202	144.76
1998	1053690	202	290.35
755	6534178	184	7649.09
631	1043035	181	307.38
1879	951590	181	435.55
1510	986912	180	365.48
1467	1085604	179	191.77
771	1068651	176	87.50
1879	1082185	166	236.96
161	868764	163	825.23
1609	1082185	160	307.64
982	6534178	153	3094.27
1707	1085604	152	169.67

Benchmarks store performance — identifies top-performing and underperforming locations

```
select
  store_id,
  round(sum(sales_value), 2) as total_revenue,
  count(distinct basket_id) as total_transactions,
  round(avg(sales_value), 2) as avg_item_value,
  round(sum(sales_value) /
    count(distinct basket_id), 2) as avg_basket_value
from transaction_data
group by store_id
order by total_revenue desc;
```

store_id	total_revenue	total_transactions	avg_item_value	avg_basket_value
Filter...	Filter...	Filter...	Filter...	Filter...
367	267614.34	7724	3.54	34.65
406	216781.77	5422	3.65	39.98
361	144698.85	5146	3.52	28.12
429	142391.01	3883	4.62	36.67
343	140455.75	5141	3.40	27.32
356	137588.68	4266	2.79	32.25
381	132488.15	5047	2.93	26.25
375	130396.51	4110	4.15	31.73
292	130005.18	3975	2.87	32.71
31782	121726.54	4496	2.97	27.07
319	117782.21	4555	3.63	25.86
321	117337.37	4366	3.25	26.88
323	116672.42	3783	3.61	30.84
32004	116392.44	4101	2.91	28.38
369	115743.39	3538	3.64	32.71
427	108389.40	3377	2.75	32.10
327	106522.19	2499	3.66	42.63

Reveals price-sensitive sub-categories where customers only buy when a coupon is available — pricing strategy red flags.

```
select
  p.sub_commodity_desc,
  count(*) total_transactions,
  sum(case when t.coupon_disc > 0
    then 1 else 0 end) coupon_transactions,
  round(sum(case when t.coupon_disc > 0
    then 1 else 0 end)* 100.0 / count(*), 2) coupon_dependency_pct
from transaction_data t
join product p on t.product_id = p.product_id
group by p.sub_commodity_desc
having count(*) > 50
order by coupon_dependency_pct desc
limit 20;
```

sub_commodity...	total_transactions	coupon_transact...	coupon_depend...
 Filter...	 Filter...	 Filter...	 Filter...
	7839	0	0.00
ABRASIVES	601	0	0.00
ACNE MEDICATIONS	488	0	0.00
ACTIVITY	144	0	0.00
ADDITIVES/FLUIDS	142	0	0.00
ADHESIVES/CAULK	190	0	0.00
ADULT ANALGESICS	3297	0	0.00
ADULT CEREAL	8829	0	0.00
ADULT INCONTINENC...	53	0	0.00
ADULT INCONTINENC...	145	0	0.00
ADULT INCONTINENC...	106	0	0.00
ADULT INCONTINENC...	226	0	0.00
ADULT PREMIUM	981	0	0.00
AEROSOL DEODORAN...	428	0	0.00
AEROSOL TOPPINGS	1177	0	0.00
AGE RESTRICTED DVD S	137	0	0.00
AGE RESTRICTED FIRE...	145	0	0.00
AIR CARE - AEROSOLS	1421	0	0.00
AIR CARE - CANDLES	749	0	0.00
AIR CARE - CONTINU...	656	0	0.00

Compares basket size and spend between weekdays and weekends to guide staffing, promotions, and restocking.

```
select
  case when (day % 7) in (0, 6) then 'Weekend'
  else 'Weekday' end as day_type,
  count(distinct basket_id) as total_baskets,
  round(sum(sales_value), 2) as total_revenue,
  round(avg(sales_value), 2) as avg_item_value,
  round(sum(sales_value) /
    count(distinct basket_id), 2) as avg_basket_value
from transaction_data
group by day_type
order by day_type;
```

day_type	total_baskets	total_revenue	avg_item_value	avg_basket_value
abc Filter...	abc Filter...	abc Filter...	abc Filter...	abc Filter...
Weekday	200920	5925777.68	3.12	29.49
Weekend	75564	2131685.40	3.07	28.21

Tracks revenue momentum — positive/negative WoW% signals campaign effectiveness or demand shifts

```
with weekly as
( select
  week_no,
  round(sum(sales_value), 2) weekly_revenue
  from transaction_data
  group by week_no )
select week_no, weekly_revenue,
  lag(weekly_revenue) over (order by week_no) prev_week_revenue,
  round((weekly_revenue - lag(weekly_revenue)
  over (order by week_no))* 100.0 / nullif(lag(weekly_revenue)
  over (order by week_no), 0), 2) wow_growth_pct
from weekly order by week_no;
```

week_no	weekly_revenue	prev_week_reve...	wow_growth_pct
abc Filter...	abc Filter...	abc Filter...	abc Filter...
1	5211.16	NULL	NULL
2	10821.35	5211.16	107.66
3	13498.20	10821.35	24.74
4	15965.99	13498.20	18.28
5	20139.82	15965.99	26.14
6	24923.90	20139.82	23.75
7	28073.97	24923.90	12.64
8	31793.77	28073.97	13.25
9	31244.17	31793.77	-1.73
10	42994.38	31244.17	37.61
11	51011.30	42994.38	18.65
12	53187.30	51011.30	4.27
13	55203.01	53187.30	3.79
14	62875.20	55203.01	13.90
15	74261.29	62875.20	18.11

Measures shopping intensity — how much customers spend and how many items they buy per trip each day

```
with basket_summary as (  
  select day, basket_id,  
    sum(quantity) as items_in_basket,  
    sum(sales_value) as basket_value  
  from transaction_data  
  group by day, basket_id )  
select day,  
  count(basket_id) as num_baskets,  
  round(avg(items_in_basket), 2) as avg_items_per_basket,  
  round(avg(basket_value), 2) as avg_basket_value  
from basket_summary  
group by day order by day;
```

day	num_baskets	avg_items_per_b...	avg_basket_value
 Filter...	 Filter...	 Filter...	 Filter...
1	16	18.88	34.33
2	13	12.00	35.30
3	44	15.41	35.46
4	44	19.30	40.58
5	25	14.72	34.28
6	37	11.57	27.25
7	34	16.41	37.39
8	43	14.05	28.00
9	39	11.92	26.95
10	51	13.67	33.62
11	60	17.32	37.77
12	60	57.20	38.45
13	48	69.08	38.82
14	72	49.74	25.50
15	62	151.18	22.12

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ROLE

Data Analyst / MIS Executive / Power BI

SKILLS

SQL · Python · Power BI · Excel

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